Lecture 6: Bending Shear Stress

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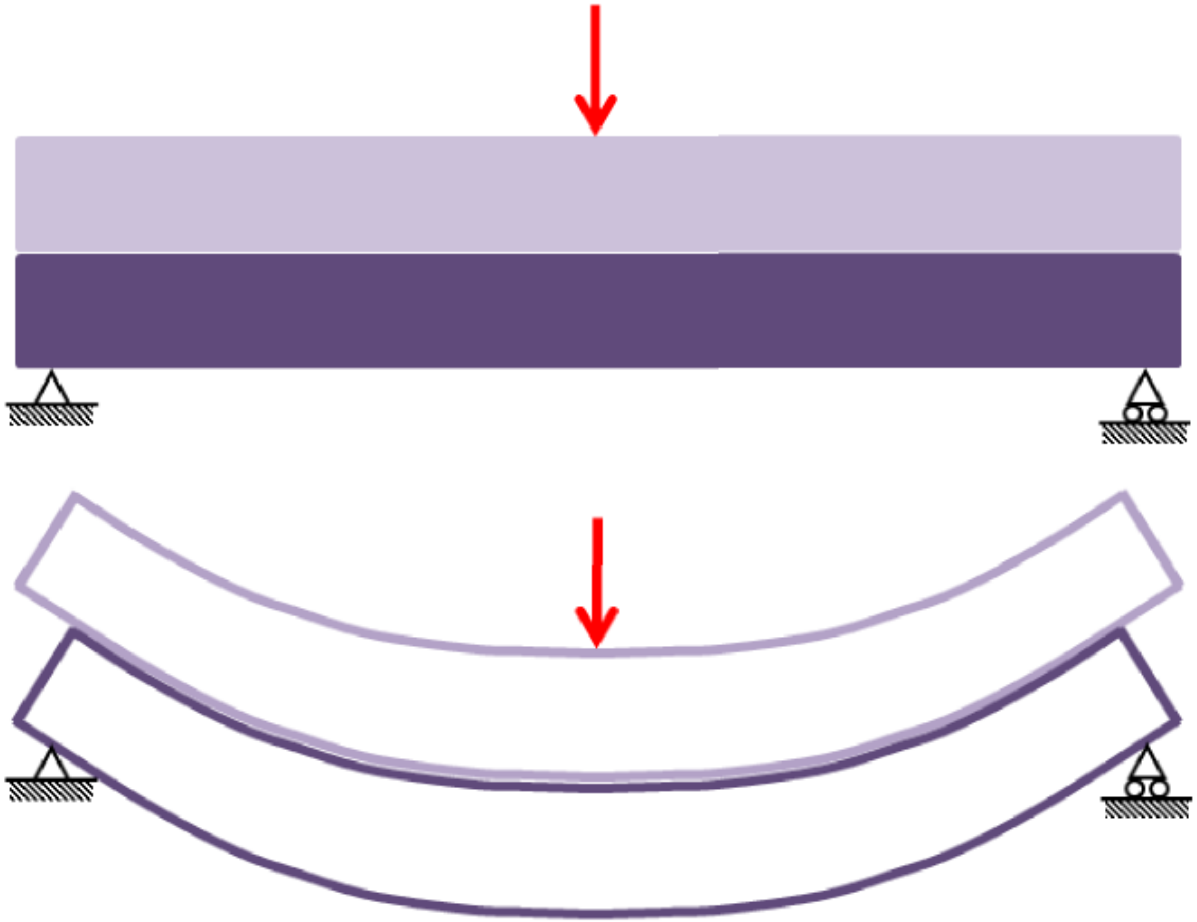
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Shear Stress

- When a beam is subjected to lateral loads ($\perp$ to the beam's longitudinal axis), both bending moment (M) and shear force (V) act on the cross section.
- The previous analysis of normal stresses in beams is for pure bending, no shear force acts on the cross sections.
- In the case of non-pure or non-uniform bending, the presence of shear forces causes warping (i.e. out of plane distortion) of the cross sections.
- Hence, a cross section that is plane before bending is no longer plane after bending.

- Although warping due to shear greatly complicates the behaviour of beams, but it has been investigated that the normal stresses (σ_x) calculated from the flexure formula are not significantly altered by the presence of shear stresses and the associated warping deformation.
- Therefore, it is justified to use the theory of pure flexure for calculating normal stresses in beams subjected to non-uniform bending.
- The existence of horizontal shear stresses in a beam can be demonstrated by bending of 2 separate beams described in the following figure:

Bending of 2 separate beams:

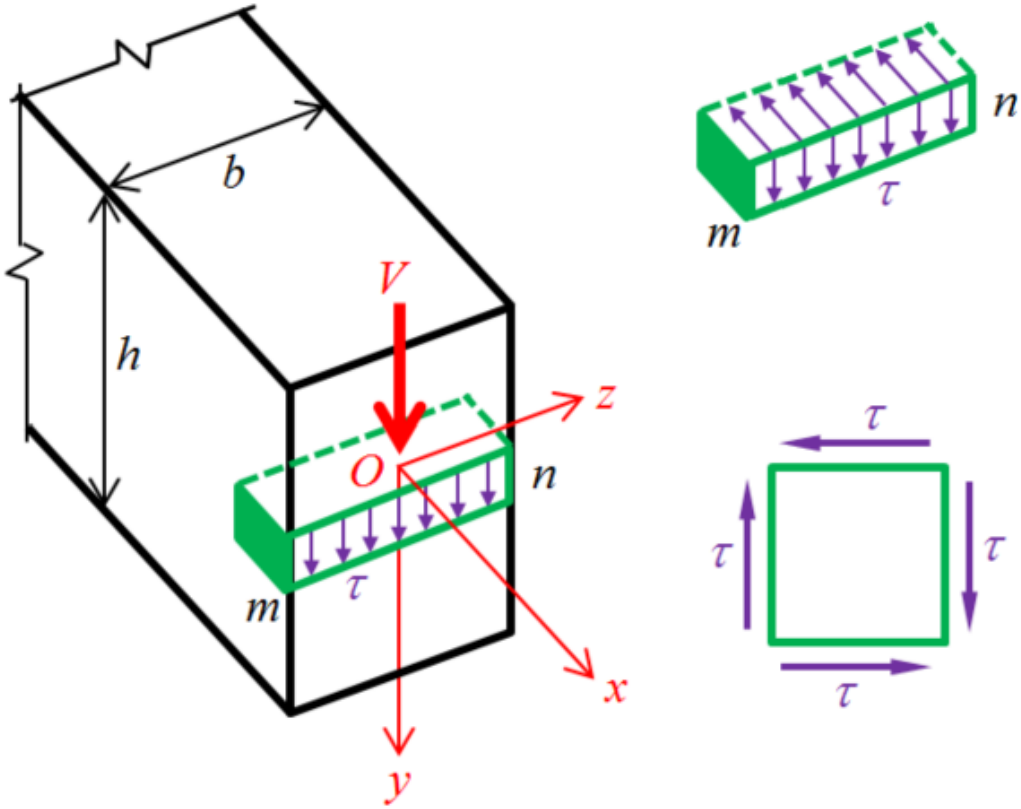


- Two equal rectangular beams are placed one on top of another and simply supported.
- There is no bond/friction between the beams.

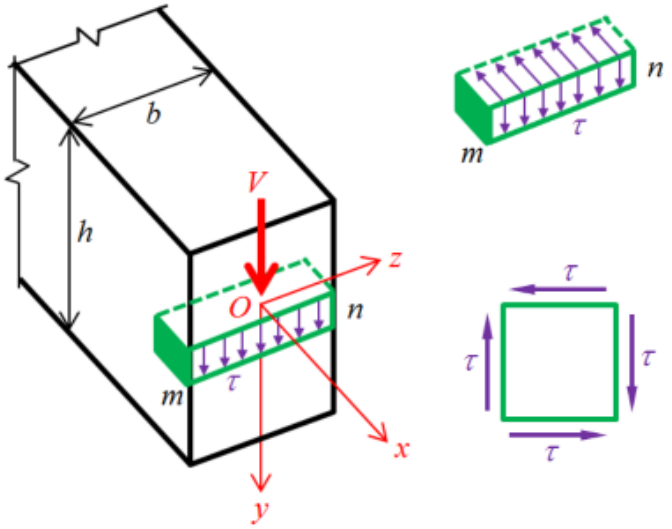
- If load is applied, bending on the 2 beams will occur independently.

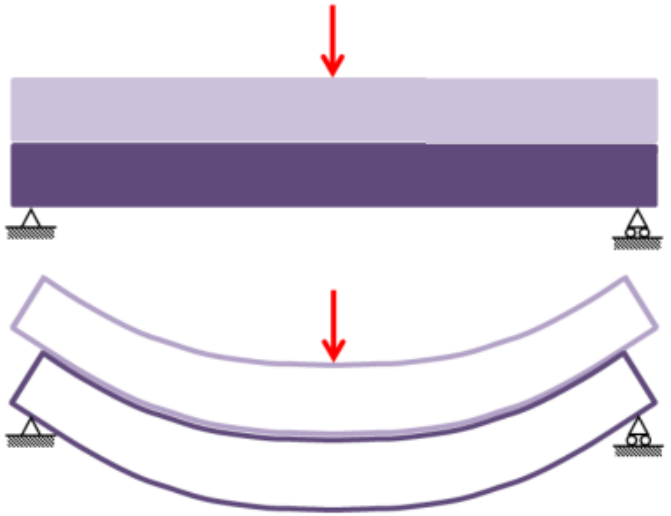
Consider a beam subjected to shear force *V*, which is vertical in direction, as follows:

- The distribution of shear stress τ at a particular level is uniform across the width of the beam.
- A small element *mn* is cut out of the beam. The vertical shear stress is accompanied by horizontal shear stress of the same magnitude.



- Hence, there will be horizontal (in longitudinal direction) shear stresses between horizontal layers in the beam.
- Remember: In a small cut out element, shear stress is always in pair and complimentary pair.
- What happens to the shear stress at the top (of the upper beam) and at the bottom (of the lower beam)?
- In fact, there are no shear stresses at either the top or the bottom of the beam.
- Hence, the vertical shear stress must also vanish at the top and bottom of the beam.

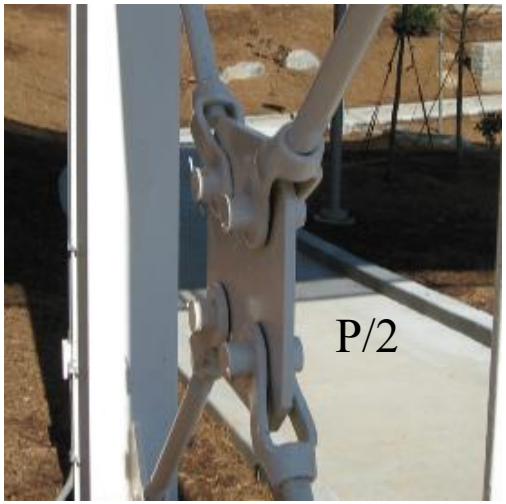




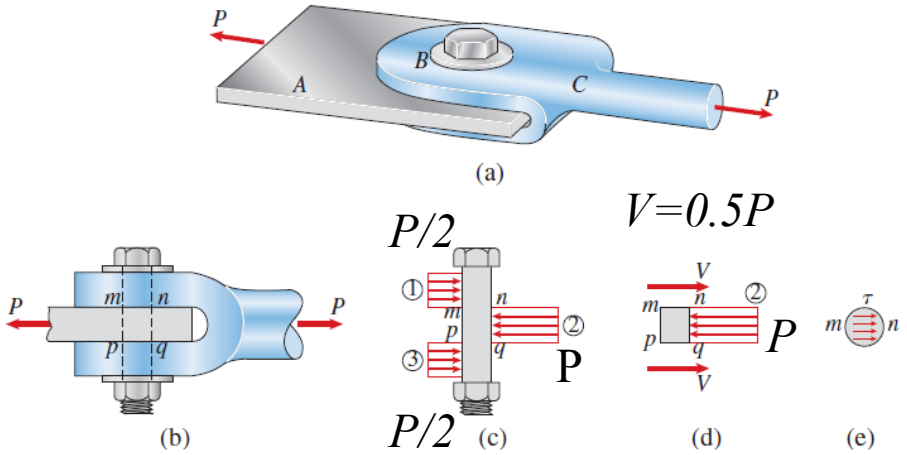
Shear Stress and Strain

Normal stress act in directions perpendicular to the surface of material. **Shear stress** act tangential to the surface of material. Consider bolted connection below. The plate A and clevis C will press B in **bearing**. The contact stress is called **bearing stress**.

$$\sigma_b = \frac{F_b}{A_b}$$



Connection of diagonal bracing



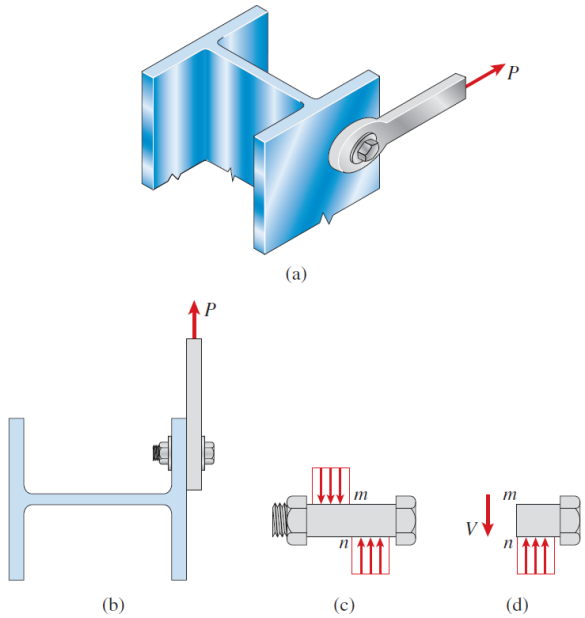
Bolted connection where bolt is under double shear

Based upon the assumption of uniform distribution, we can calculate an average bearing stress σ_b by dividing the total bearing force Fb by the bearing area Ab. Bearing force at region 2 is equivalent to the axial tensile force P. Bearing force at region 1 is 0.5P. we see that shear forces V act over the cut surfaces of the bolt. There are two planes of shear (*mn* and *pq*), and so the bolt is said to be in double shear. In double shear, each of the shear forces is equal to one-half of the total load transmitted by the bolt, that is, V=P/2. The **average shear stress** on surface *mn* is:

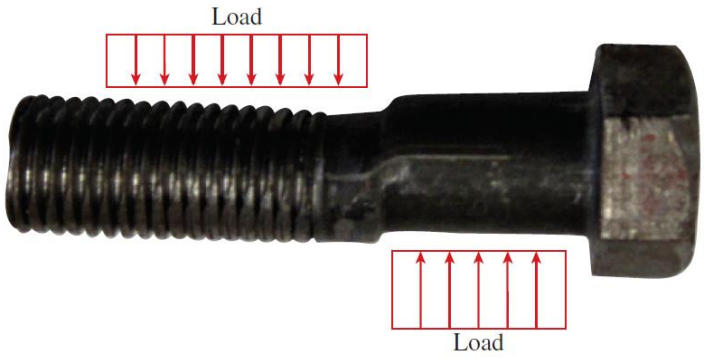
$$\tau_{aver} = \frac{V}{A}$$

Simple shear (direct shear)

The following figure shows single shear, where the shear force applied to the stud is P



Bolted connection where bolt is in single shear



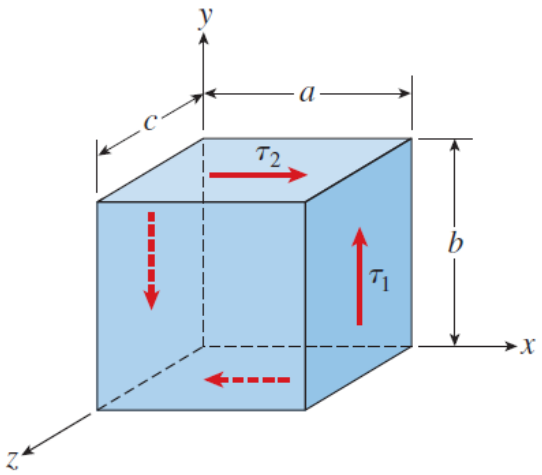
Failure of stud due to shear force and shear stress

Note: the example in this page and previous page are examples of direct shear (or simple shear) in which the shear stresses are created by the direct action of the forces in trying to cut through the material. Direct shear arises in the design of bolts, pins, rivets, keys, welds, and glued joints. Shear stresses also arise in an indirect manner when members are subjected to tension, torsion, and bending,

$$\tau_{\text{aver}} = \frac{V}{A}$$

Equality of shear stress on perpendicular planes

1. Consider a small element of material in the form of a rectangular parallelepiped having sides of lengths a, b, and c in the x, y, and z directions, respectively. Now assume that a shear stress τ_1 is distributed uniformly over the right-hand face, which has area bc. In order for the element to be in equilibrium in the y direction, the total shear force $\tau_1$bc acting on the righthand face must be balanced by an equal but oppositely directed shear force on the left-hand face. Since the areas of these two faces are equal, it follows that the shear stresses on the two faces must be equal.



Small element of material subjected to shear stresses

$$\tau_1 = \tau_2$$

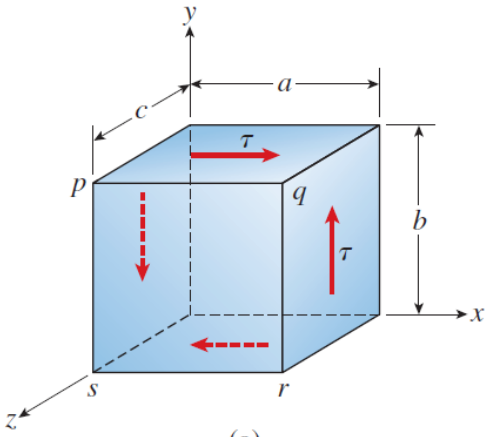
2. The forces $\tau_1 bc$ acting on the left- and right-hand side faces form a couple having a moment about the z axis of magnitude $\tau 1abc$, acting counterclockwise in the figure. Equilibrium of the element requires that this moment be balanced by an equal and opposite moment resulting from shear stresses acting on the top and bottom faces of the element. Denoting the stresses on the top and bottom faces as τ_2, we see that the corresponding horizontal shear forces equal $\tau_2 ac$. These forces form a clockwise couple of moment $\tau_2 abc$. From moment equilibrium of the element about the z axis, we see that $\tau_1 abc = \tau_2 abc$

Assumption: Assume this is very small segment and external force (such as gravity) are negligible.

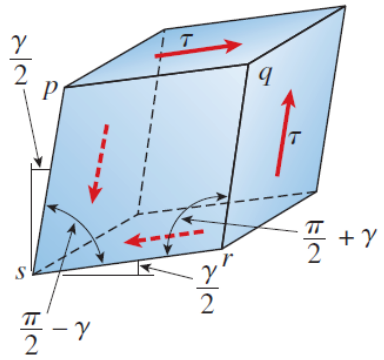
Equality of shear stress on perpendicular planes

Conclusion 1. Shear stresses on opposite (and parallel) faces of an element are equal in magnitude and opposite in direction.

Conclusion 2. Shear stresses on adjacent (and perpendicular) faces of an element are equal in magnitude and have directions such that both stresses point toward, or both point away from, the line of intersection of the faces.



(a)



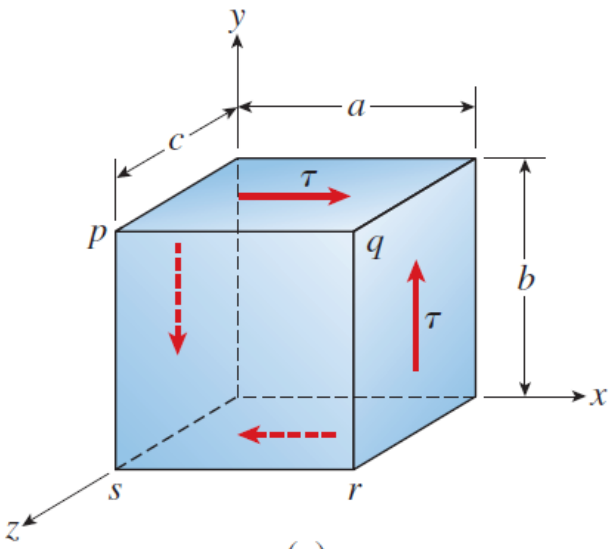
(b)

This conclusions 1 and 2 are derived from **pure shear** stress state (no normal stress on these surfaces).

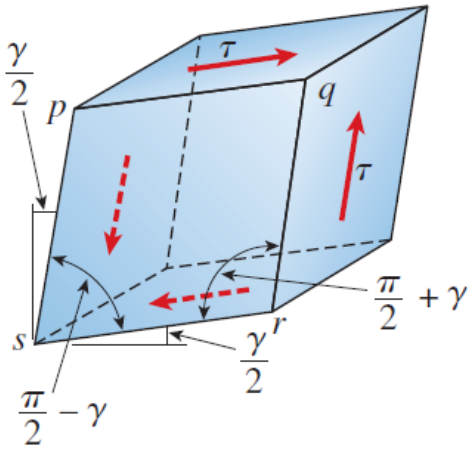
When there are uniform axial stress field added, both conclusions 1 and 2 hold.
When non-uniform axial stress filed is added, if the axial stress change too much, conclusion 2 is still correct. [see the derivation later in this lecture note for shear stress distribution]

Shear stress are accompanied by shear strains. Shear stress has no tendency to elongate or shorten in x,y,z axis. The length of each side will not change, shear stress only produce change in the shape of element. The rectangle turn to parallelepiped.

Shear strain



(a)

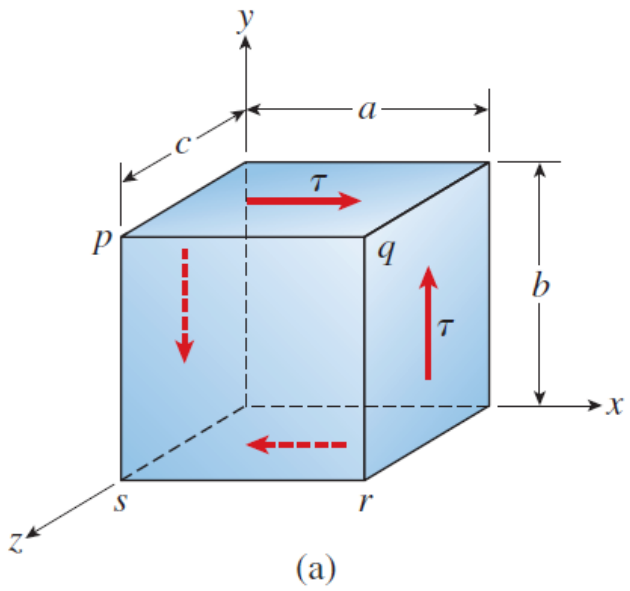


(b)

Because of this deformation, the angles between the side faces change. For instance, the angles at points q and s, which were $\pi/2$ before deformation, are reduced by a small angle γ to $\pi/2$-γ. The angles at points p and r are increased to π /2+γ. The angle γ is a measure of the **distortion**, or change in shape, of the element and is called the **engineering shear strain** (or shear strain, in this course). 0.5γ is **scientific shear strain**. Because shear strain is an angle, it is usually measured in degrees or radians.

Shear strain in an element is positive when the angle between two positive faces (or two negative faces) is reduced. The strain is negative when the angle between two positive (or two negative) faces is increased.

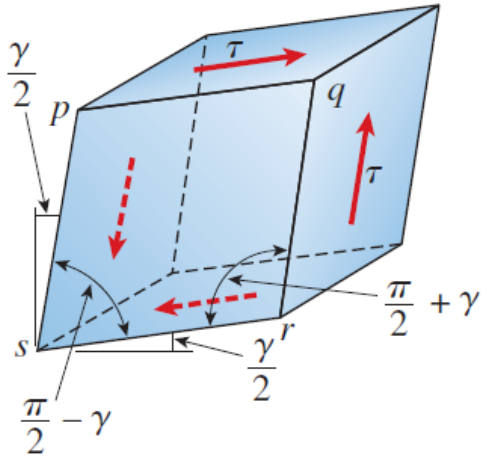
Sign convention of shear stress and strain



(a)

A shear stress acting on a positive face of an element is positive if it acts in the positive direction of one of the coordinate axes and negative if it acts in the negative direction of an axis.

A shear stress acting on a negative face of an element is positive if it acts in the negative direction of an axis and negative if it acts in a positive direction.



(b)

Shear strain in an element is positive when the angle between two positive faces (or two negative faces) is reduced. The strain is negative when the angle between two positive (or two negative) faces is increased.

For most material, shear modulus is positive in elastic region, therefore shear strain and shear stress have same sign for elastic material.

Hooke's law in shear

For material in pure shear loading, we can plot the shear stress-strain diagram based on material coupon test. The initial part of shear stress-strain diagram for most material has linear elastic region. Therefore, it follow the Hooke's law in shear.

$$\tau = G\gamma$$

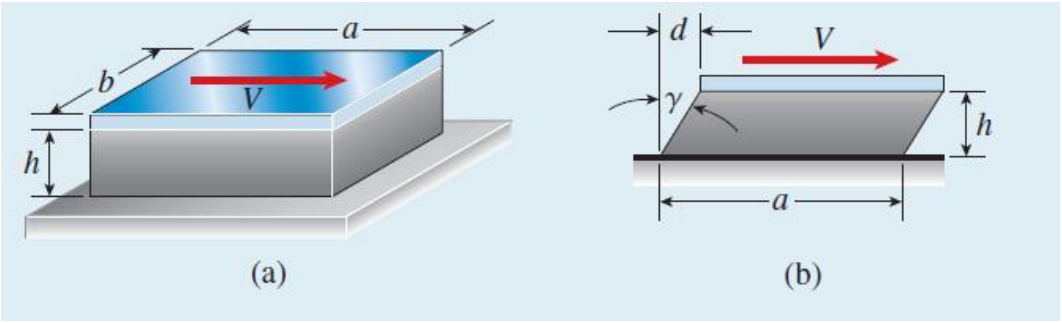
G is the shear modulus of elasticity (also called the *modulus of rigidity*).

The shear modulus *G* has the same units as the tension modulus *E*, namely, pascals (or multiples thereof) in SI units. For mild steel, typical values of *G* are 75 GPa; for aluminum alloys, typical values are 28 GPa.

$$G = \frac{E}{2(1 + \nu)}$$

Example 5.1: rubber bearing pad

A bearing pad of the kind used to support machines and bridge girders consists of a linearly elastic material (usually an elastomer, such as rubber) capped by a steel plate. Assume that the thickness of the elastomer is h, the dimensions of the plate are a b, and the pad is subjected to a horizontal shear force V. The rubber Shear modulus is G_e. Find horizontal displacement d



$$\tau_{aver} = \frac{V}{ab}$$

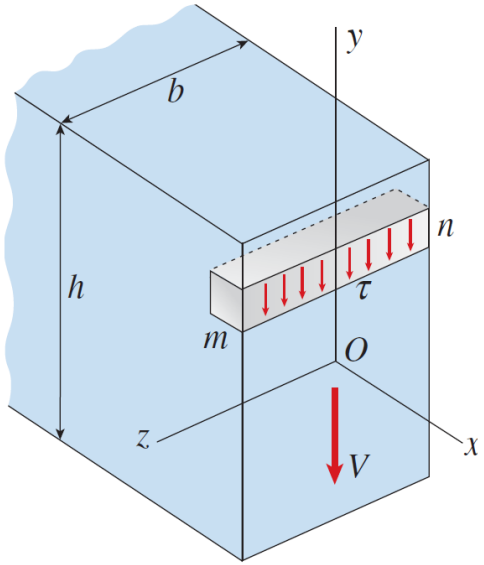
$$\gamma = \frac{\tau_{aver}}{G_e} = \frac{V}{abG_e}$$

$$d = h\tan\gamma = h\tan\left(\frac{V}{abG_e}\right)$$

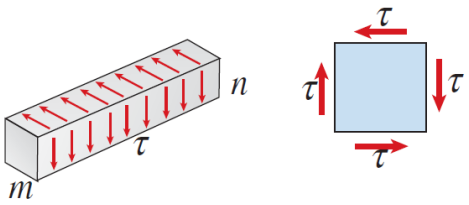
Assume displacement is small: Taylor expansion

$$d = h\gamma = \frac{hV}{abG_e}$$

Shear stress in beams of rectangular section



(a)

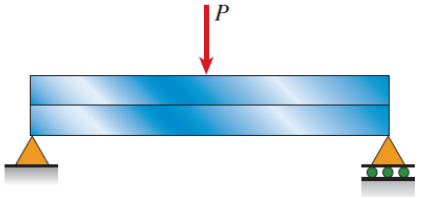


Consider a small body mn cut out from the beam. The shear stress is assumed to be uniform along the z coordinate. But the shear force may vary by height. Cut a small segment.

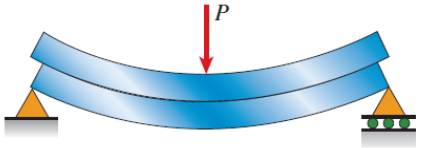
At the top fiber and bottom fiber of beam, there is no horizontal shear stress, therefore, also no vertical shear stress.

At other location, horizontal and vertical shear stress happen and are equal [proved before].

If no horizontal shear stress, at middle height, the beam will become two separate beams as below.

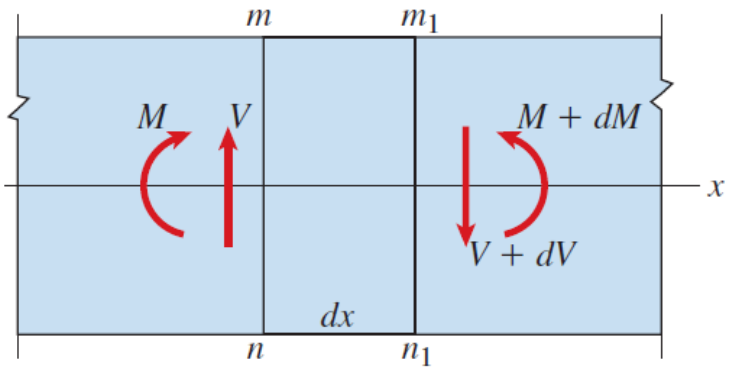


(a)

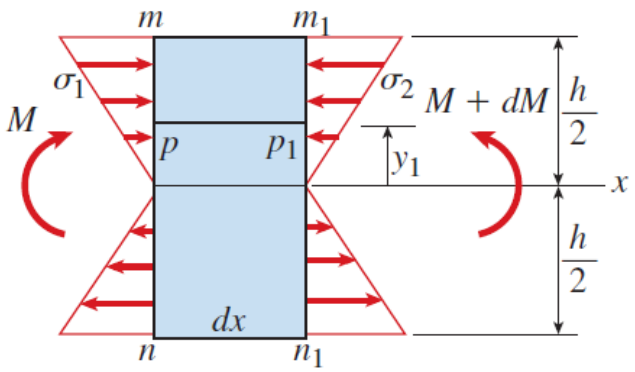


Shear stress in beams with rectangular section

Derivation of shear stress formula

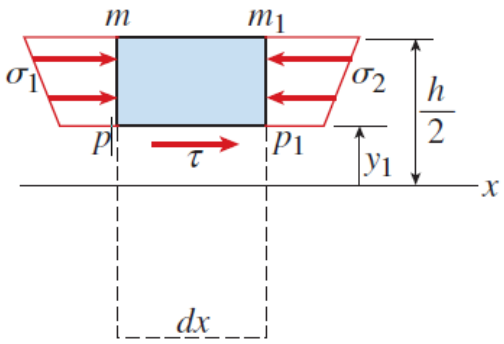


Side view of beam
(a)

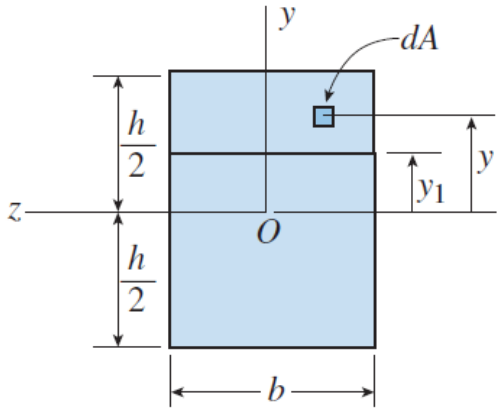


Side view of element
(b)

Assumption: linear elastic material under combination of all loads. Small deformation.



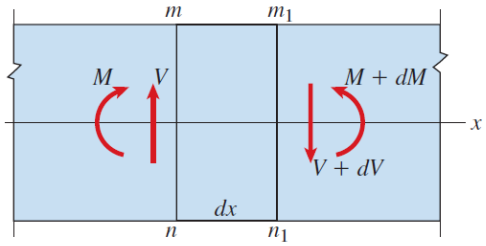
Side view of subelement
(c)



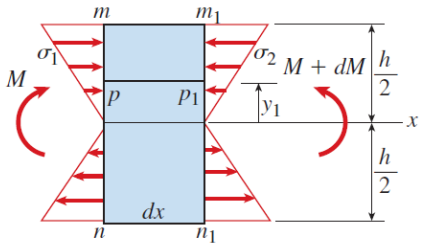
Cross section of beam at subelement
(d)

Shear stress development in a beam due to shear force and moment
Here we neglect axial force. Even if axial force exist, it will not affect solution for shear stress

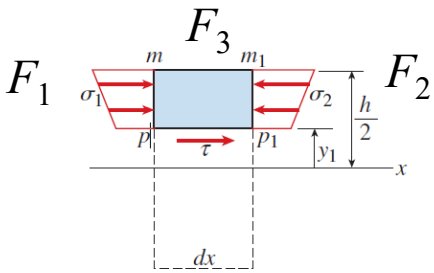
Derivation of shear stress formula



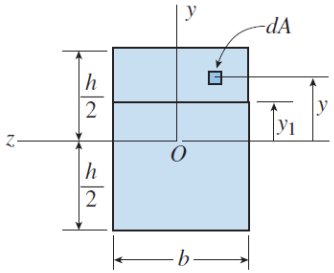
Side view of beam
(a)



Side view of element
(b)



Side view of subelement
(c)



Cross section of beam at subelement
(d)

Instead of deriving the vertical shear stress on m_1-p_1, we derive the horizontal shear stress on p-p1 surface. Obtain the normal stress distribution on two surfaces.

$$\sigma_1 = -\frac{My}{I} \quad \text{and} \quad \sigma_2 = -\frac{(M + dM)y}{I}$$

$$\sigma_1 dA = \frac{My}{I} dA$$

$$F_1 = \int \sigma_1 \, dA = \int \frac{My}{I} dA$$

$$F_2 = \int \sigma_2 dA = \int \frac{(M + dM)y}{I} dA$$

$$F_3 = F_2 - F_1$$

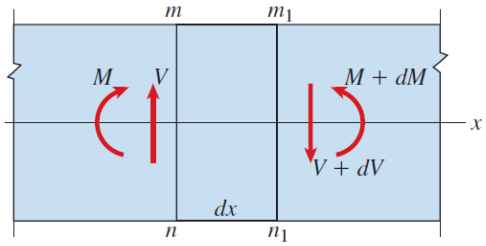
Since the subelement is in equilibrium, we can sum forces in the *x* direction and obtain:

$$F_3 = \tau b \, dx$$

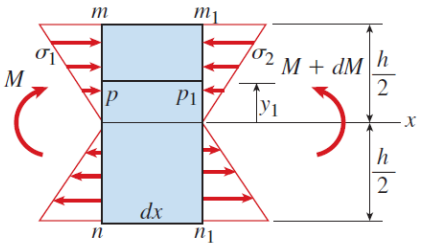
$$F_3 = \int \frac{(M + dM)y}{I} dA - \int \frac{My}{I} dA = \int \frac{(dM)y}{I} dA$$

$$\tau = \frac{dM}{dx}\left(\frac{1}{Ib}\right)\int y \, dA \qquad \tau = \frac{V}{Ib}\int y dA \qquad Q(y_1) = \int_{y_1}^{\text{top}} y dA \qquad \tau(y_1) = \frac{VQ(y_1)}{Ib}$$

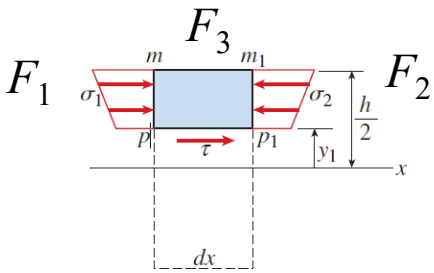
Derivation of shear formula



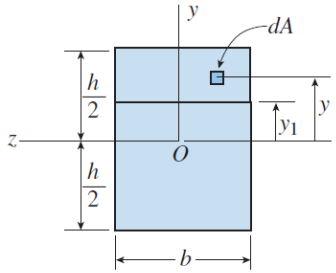
Side view of beam
(a)



Side view of element
(b)



Side view of subelement
(c)



Cross section of beam at subelement
(d)

Shear formula

$$\tau(y_1) = \frac{VQ(y_1)}{Ib}$$

$$Q(y_1) = \int_{y_1}^{\text{top}} y\,dA$$

$$\tau = \frac{VQ}{Ib}$$

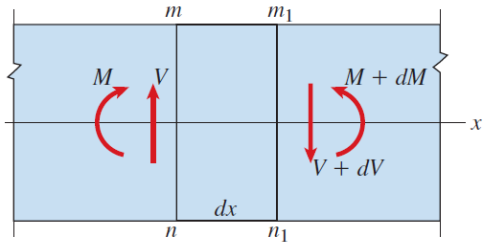
$Q(y_1)$: First moment of the cross-sectional area above y_1, the level at which shear stress is evaluated.

Assumption: linear elastic material under combination of all loads. Small deformation.

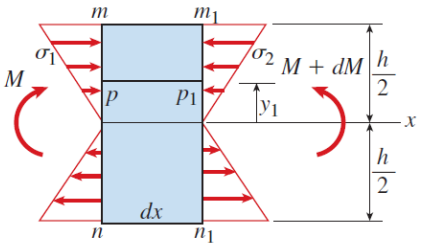
We could also calculate the first moment of the remaining cross-sectional area, that is, the area *below* the shaded area. Its first moment is equal to the negative of Q.
If you integrate the whole cross section, $Q(y_1=-0.5h)=0$.

No need to bother with sign conventions for *V* and *Q*. Instead, we treat all terms in the shear formula as positive quantities and determine the direction of the shear stresses by inspection. Shear stress flow always in the direction of shear force *V*.

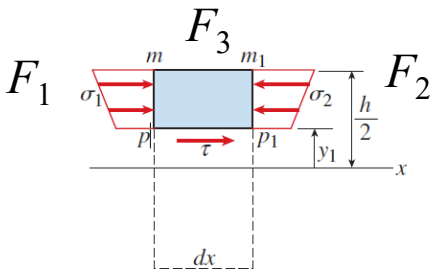
Distribution of shear stress in rectangular beam



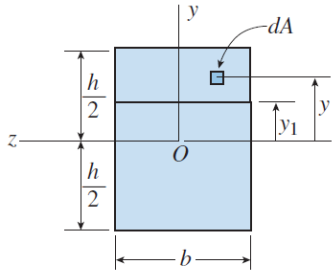
Side view of beam
(a)



Side view of element
(b)

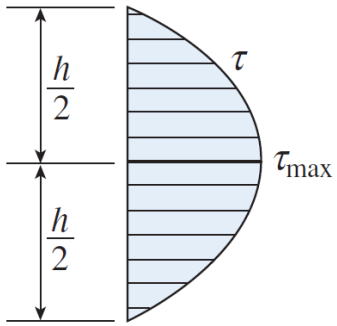


Side view of subelement
(c)



Cross section of beam at subelement
(d)

Note: the shear stress distribution is a parabolic curve over the beam height for rectangular beam.



$$Q = \int y dA$$

Choice 1: Area multiplied by distance from neutral axis to center

$$Q = b\left(\frac{h}{2} - y_1\right)\left(y_1 + \frac{h/2 - y_1}{2}\right) = \frac{b}{2}\left(\frac{h^2}{4} - y_1^2\right)$$

Choice 2: Write the integration explicitly

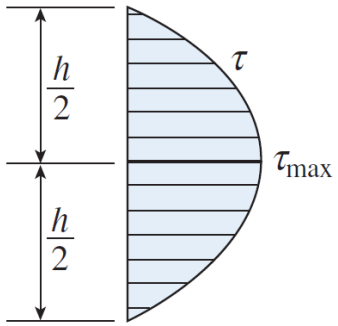
$$Q = \int y\, dA = \int_{y_1}^{h/2} yb\, dy = \frac{b}{2}\left(\frac{h^2}{4} - y_1^2\right)$$

$$\tau = \frac{V}{2I}\left(\frac{h^2}{4} - y_1^2\right)$$

$$\tau_{\max} = \frac{Vh^2}{8I} = \frac{3V}{2A}$$

Assumptions for rectangular beam: (1) The edges of the cross section must be parallel to the y axis (so that the shear stresses act parallel to the y axis), and (2) the shear stresses must be uniform across the width of the cross section.

Distribution of shear stress in rectangular beam

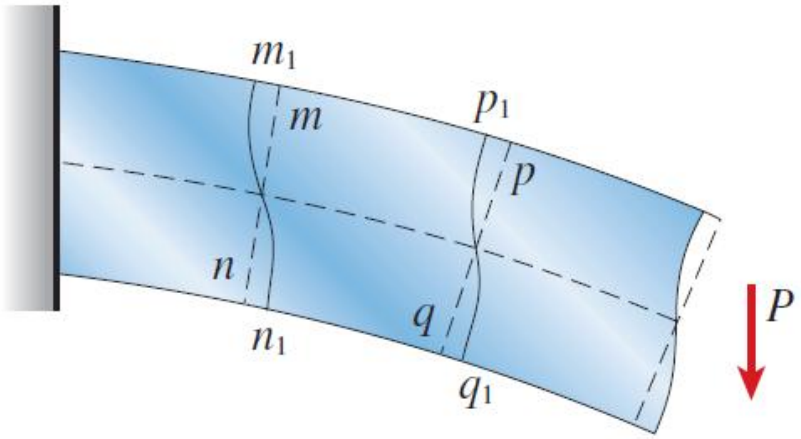


$$\tau = \frac{V}{2I}\left(\frac{h^2}{4} - y_1^2\right)$$

$$\tau_{max} = \frac{Vh^2}{8I} = \frac{3V}{2A}$$

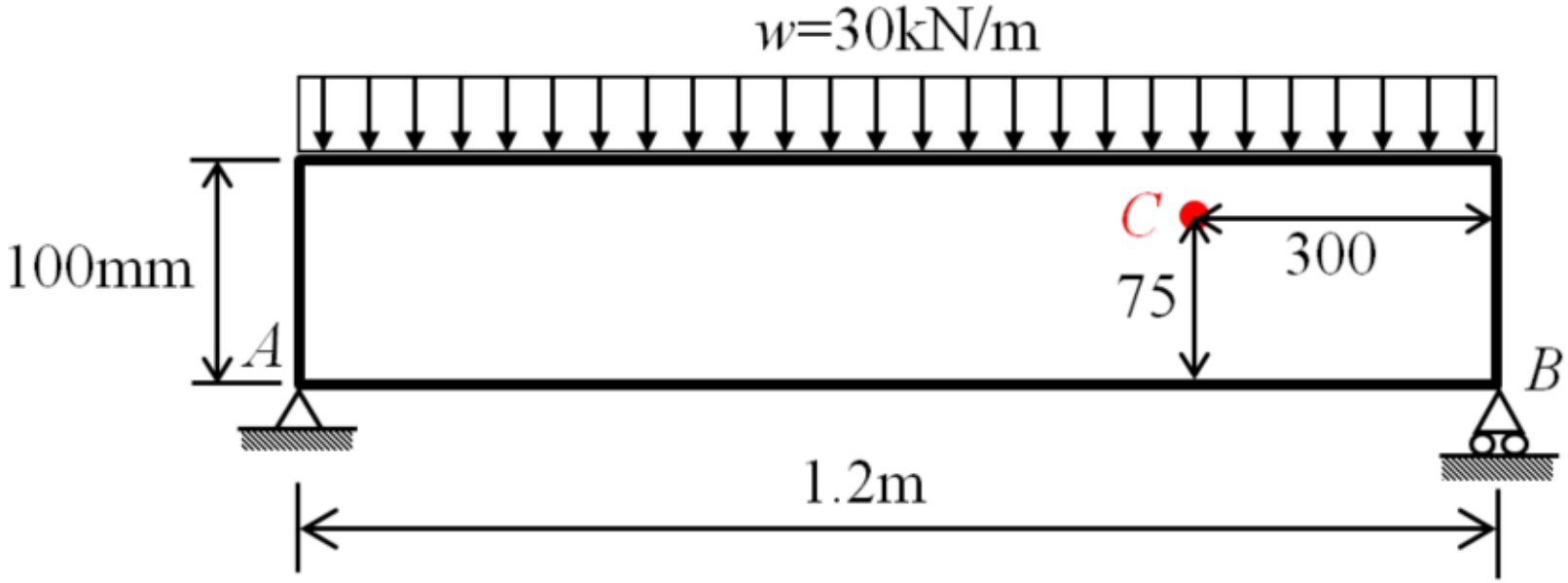
The maximum shear stress is 50% higher than average shear stress for this solution. The vertical shear stress is equal to the horizontal shear stress.

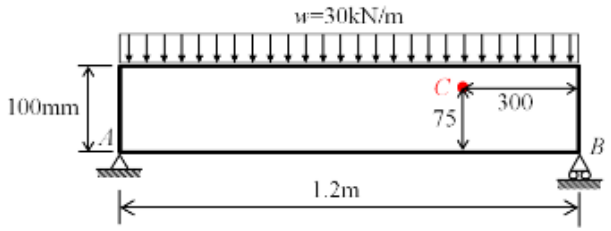
The shear stress is parabolic so shear strain is also parabolic. Here the shear strain is negative based on previous definition. So the shear deformation is similar to this result:
Warping: horizontal shear stress caused plane section to no longer remain plane section (the warping deformation is small).

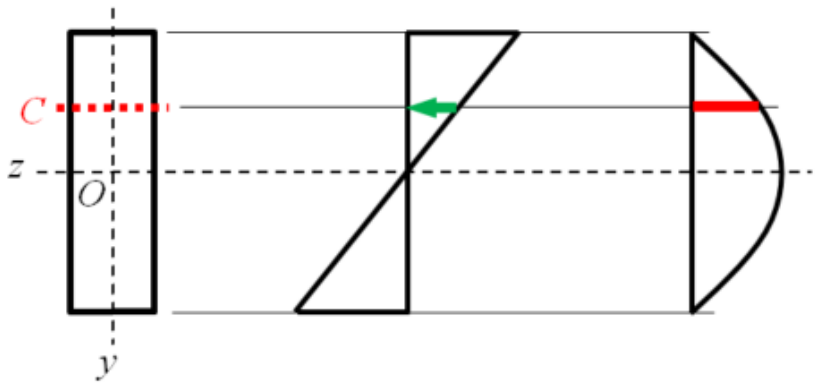


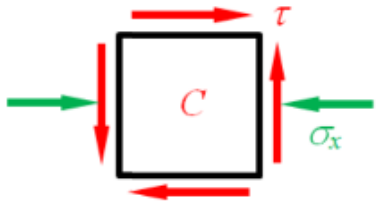
Example 2:

- Calculate the magnitude of normal stress and shear stress acting at element *C*, which is located 75mm above and 300mm to the left of support *B*.
- Determine their directions.
- The span and cross-section dimensions of the beam are respectively 1.2m and 30×100mm (*b*×*h*).









Cross section | Normal stress σ_x | Shear stress τ | Shear stress direction

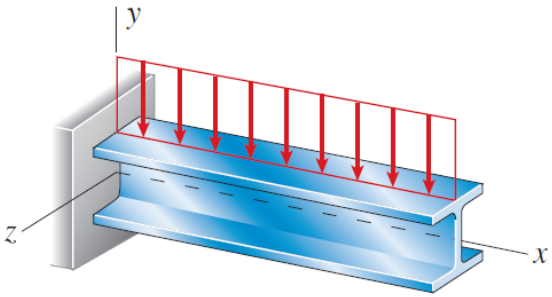
Answer:

- Reactions at A & B: $R_A = R_B = R = ½\times30\times1.2 = 18\text{kN}$
- BM at C: $M_C = 18\times0.3 - ½\times30\times0.3^2 = 4.05\text{kNm}$
- Moment of inertia: $\frac{1}{12} \times 30 \times 100^3 = 2.5 \times 10^6\ \text{mm}^4$
- $\therefore$ Normal stress at C: $\sigma = \frac{My}{I} = \frac{4.05\times0^6\times(-25)}{2.5\times10^6} =$

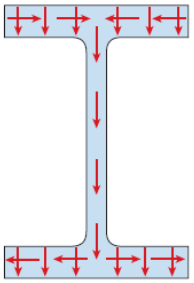
  -40.5MPa (compression)
- Shear force at C: $V_C = R - wx = 18 - 30\times0.3 = 9\text{kN}$
- $\therefore$ Shear stress at C: $\tau = \frac{VQ}{bI} = \frac{9000\times(25\times30\times37.5)}{30\times2.5\times10^6} =$

  3.4 MPa (for direction, see the element)

Distribution of shear stress in webs of I-section beam



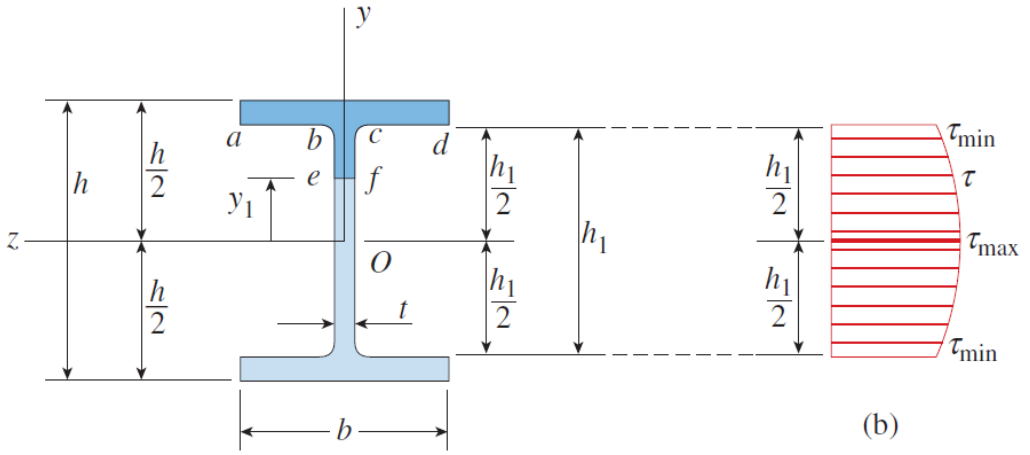
(a)



(b)

The distribution of the shear stresses in a wide-flange beam is more complicated than in a rectangular beam. For instance, the shear stresses in the flanges of the beam act in both vertical and horizontal directions. Horizontal shear stress are much greater than vertical shear stress in the flange.

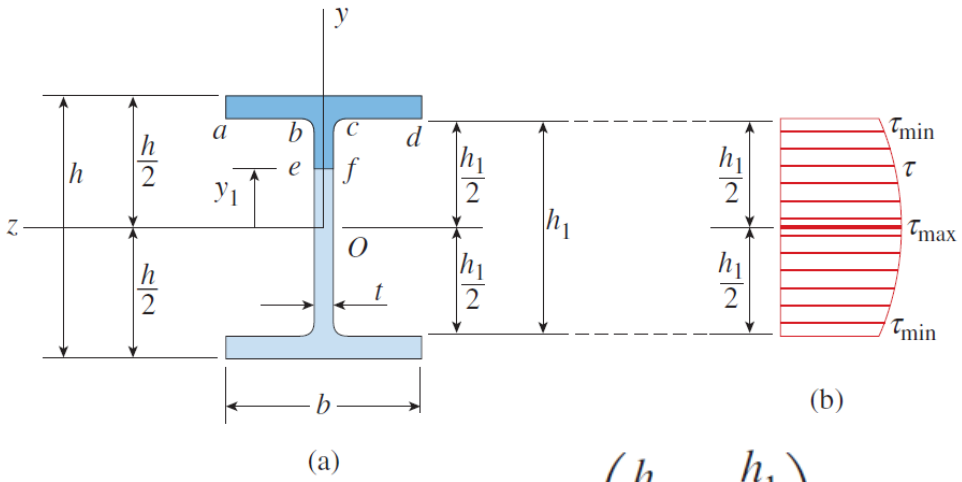
The shear stresses in the web of a wide-flange beam act only in the vertical direction and are larger than the stresses in the flanges. These stresses can be found by the same techniques we used for finding shear stresses in rectangular beams.



Assumptions for rectangular beam still hold here: (1) The edges of the cross section must be parallel to the y axis (so that the shear stresses act parallel to the y axis), and (2) the shear stresses must be uniform across the width of the cross section.

Distribution of shear stress in webs of I-section beam

In this page, only y_1 is unknown, the rest values (t, h, h_1 are all known)



A_1: area in the top flange

$$A_1 = b\left(\frac{h}{2} - \frac{h_1}{2}\right)$$

A_2: area in the web

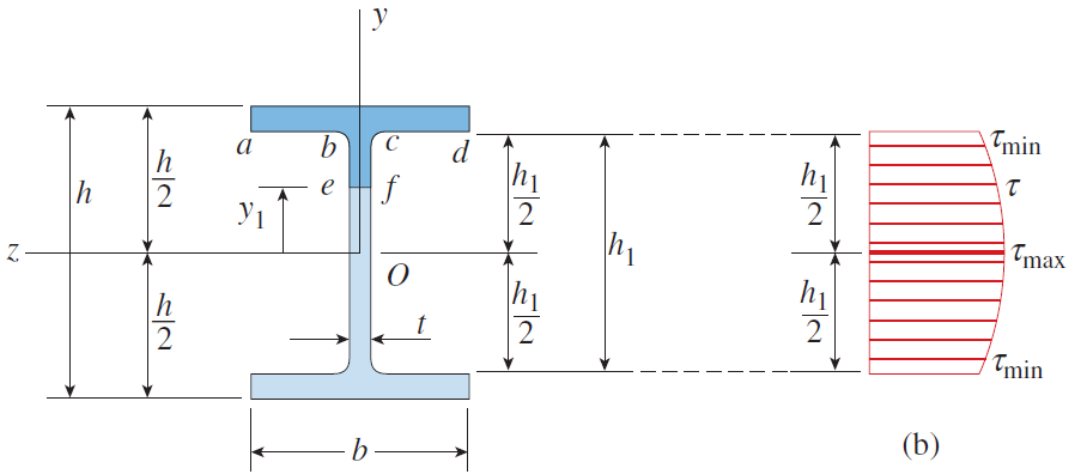
$$A_2 = t\left(\frac{h_1}{2} - y_1\right)$$

$$Q = \frac{b}{8}(h^2 - h_1^2) + \frac{t}{8}(h_1^2 - 4y_1^2) \qquad \tau = \frac{VQ}{It} = \frac{V}{8It}\left[b(h^2 - h_1^2) + t(h_1^2 - 4y_1^2)\right]$$

Quadratic function of y_1

$$I = \frac{bh^3}{12} - \frac{(b-t)h_1^3}{12} = \frac{1}{12}(bh^3 - bh_1^3 + th_1^3)$$

Distribution of shear stress in webs of I-section beam



$$\tau = \frac{VQ}{It} = \frac{V}{8It}\left[b(h^2 - h_1^2) + t(h_1^2 - 4y_1^2)\right]$$

Maximum and minimum shear stress in web:

$$\tau_{max} = \frac{V}{8It}(bh^2 - bh_1^2 + th_1^2) \qquad \tau_{min} = \frac{Vb}{8It}(h^2 - h_1^2)$$

Total web shear force:

$$V_{web} = \frac{th_1}{3}(2\tau_{max} + \tau_{min})$$

For most engineering application of I-beams, V_{web} is more than 90% of total shear force. It is reasonable to divide the total shear force V by the area of web to obtain average shear stress of web for design purpose (assume web stress is constant). That is $\tau = V/th_1$

Summary and Review

1. The shear stress and shear strain are both defined. The **positive direction** of shear stress and shear strain are summarized.
2. The vertical shear stress and the horizontal shear stress are **proved to be equal** based on moment equilibrium equation of tiny free body cut from member.
3. The normal stresses calculated from the **flexure formula** are not significantly altered by the presence of **shear stresses and the associated warping** of the cross section for the case of nonuniform bending.
4. Beams subject to bending moment and shear force develop both normal and shear stress. Normal stress are calculated from **flexure formula** and shear stress are calculated from **shear formula**.

$$\tau = \frac{VQ}{Ib}$$

Summary and Review

5. The maximum shear stress (and maximum shear strain) happen at the neutral axis and the shear stress and shear strain are 0 at top and bottom surface of the beam.
6. The shear formula only applies only to prismatic beams and is valid only for beams of linearly elastic materials with small deflections; also, the edges of the cross section must be parallel to the y axis. For rectangular cross sections:

$$\tau_{\max} = \frac{3}{2}\frac{V}{A}$$

7. For I-shaped section, it is reasonable to assume the shear stress is uniformly distributed in the web plate, where t is thickness of web and h_1 is height of web:

$$\tau = \frac{V}{th_1}$$

8. The shear stress rarely govern the design of steel or aluminum beams. For wood beams or concrete beams, shear may govern because these materials are weak in shear.

Summary and Review

9. Shear stresses in the flanges of wide-flange beams act in both vertical and horizontal directions.
10. The shear stresses in the web of a wide-flange beam act only in the vertical direction and may be computed using the shear formula.
11. The maximum shear stress in the web of a wide-flange beam occurs at the neutral axis, and the minimum shear stress occurs where the web meets the flanges.